

A 3 bit 36 GS/s Flash ADC in 65 nm Low Power CMOS Technology

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Abstract—A 36 GS/s 3 bit flash ADC with a large analog input bandwidth is realized in a 65 nm CMOS technology. By employing a fourfold parallelization a high sampling rate is achieved, while a large input bandwidth is maintained. The measured effective resolution is about 2 bit up to 20 GHz input signal frequency at a sampling rate of 36 GS/s. The power consumption of the ADC core is 2.6 W, the core area is 0.16 mm². The ADC is intended for the cost-effective integration with an equalizer circuit on a single CMOS chip.

I. INTRODUCTION

The dispersion in fiber optical systems limits the maximum transmission distance in 10 to 40 Gbit/s fiber-optic systems. A cost effective solution for this problem is the electronic dispersion compensation by means of a digital equalization circuit. This solution requires a high speed analog-to-digital converter (ADC) with a minimum sampling rate that is equal to the data transmission rate. For on-off keying transmission systems a relatively low nominal resolution of about 3 bit is sufficient [1], [2].

The advances in CMOS technology allow the design of analog circuits in the range of multiple Gigahertz, which was traditionally dominated by bipolar technologies. Due to the high integration in CMOS technology and the fast transistors expensive multi-chip solutions can be replaced by a cost-efficient single chip CMOS solution.

The fastest ADC available today is a 8 bit 56 GS/s CMOS ADC [3]. Although the performance of this ADC is higher, the chip area is more than 100 times larger than the chip area of the 3 bit ADC presented here.

The simulation results of the ADC were presented in [4] and the first measurement results at 12.8 GS/s and 18 GS/s were presented in [5]. The layout and the circuit are also covered in detail in [4], [5]. In this paper the linearity measurement and the SNDR measurement results up to 36 GS/s are presented.

The micrograph of the chip is depicted in Fig. 1. With a core area of only 0.16 mm², this ADC is the smallest compared to any state-of-the-art ADCs with sampling rates in this range. This allows for a cost-effective integration of multiple converters on a single chip. In the above described application of a 40 Gbit/s equalizer the ADC can be integrated together with an equalizer on a single chip in a state-of-the-art CMOS technology to realize a cost-effective data transmission system [6].

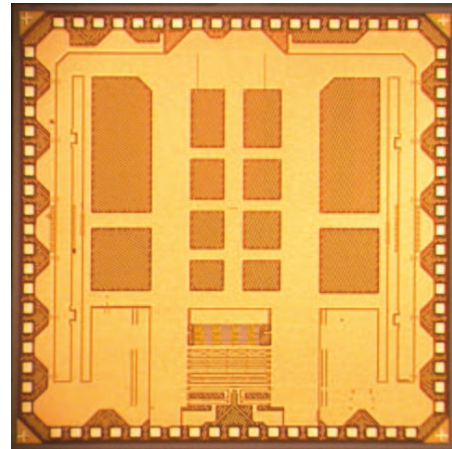


Fig. 1. Micrograph of the ADC (2.3 x 2.3 mm²).

II. ADC ARCHITECTURE

The basic concept of the ADC is a fourfold parallelized flash ADC which is fed by a four phase clock divider (Fig. 2). The analog input is demultiplexed by the sample and hold circuits. Following the sample and hold circuits quantizers and decoders are performing the quantization of the sampled input signal. Subsequently the thermometer code is encoded into the binary code. Finally all four channels are synchronized by a three step synchronizer to allow for a synchronous data output of all four channels. The detailed description of the building blocks is covered in [4], [5].

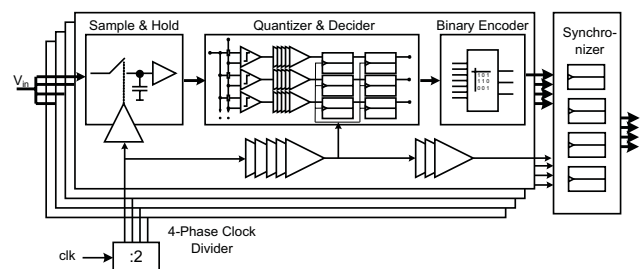


Fig. 2. Block diagram of the 3 bit ADC.

III. MEASUREMENT RESULTS

The ADC chip is mounted on Taconic RF-60A substrate for the measurements. The chip is placed in the center of the substrate and bonded to microstrip transmission lines. The layout has a star-like appearance with SMP connectors that are arranged in about 6 cm distance around the chip.

In the first measurement the DC characteristics are measured and the integral non-linearity (INL) and the differential non-linearity (DNL) is calculated. In the second measurement the signal to noise and distortion ratio (SNDR) is measured up to a total sampling rate of 36 GS/s.

A. DC Characteristics

The measurement setup for the linearity measurement is shown in Fig. 3. A Parameter Analyzer is used to generated the static analog input voltages. The ADC output data is stored in a Virtex4 FPGA and read out by a PC where the data is evaluated.

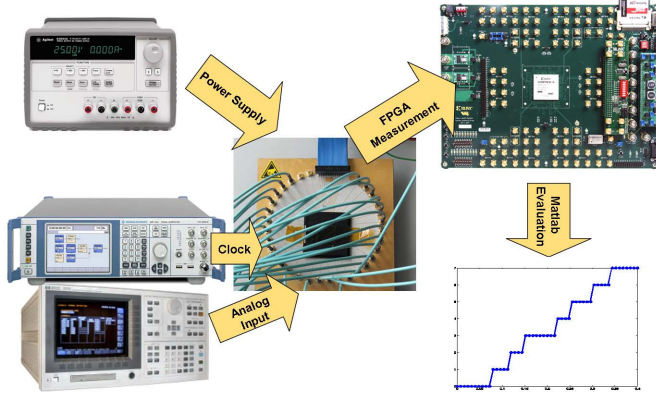


Fig. 3. Measurement setup for the INL and DNL measurement.

Fig. 4 shows the INL plot. The INL of channel 3 is 0.3 LSB, the INL of the other channels is 0.4 LSB. The maximum mismatch between the channels is 1.2 LSB.

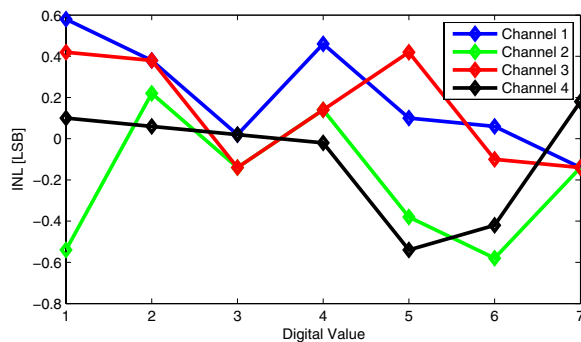


Fig. 4. The INL of the four ADC channels.

The DNL plots are depicted in Fig. 5. The DNL of the channels 1 to 4 is 0.4 LSB, 0.8 LSB, 0.6 LSB and 0.6 LSB.

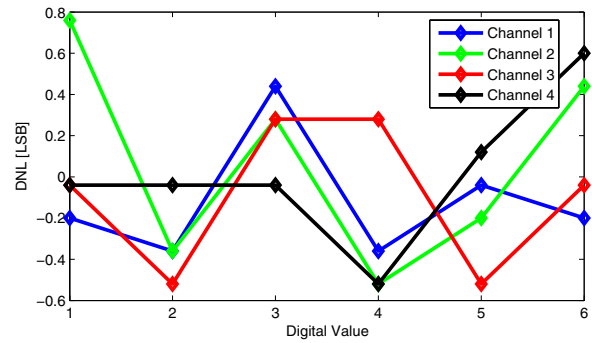


Fig. 5. The DNL of the four ADC channels.

B. SNDR Measurement

For the SNDR measurement the setup in Fig. 3 is slightly modified. The input signal is generated by a sinusoid signal generator and balanced with a 180° -hybrid. The amplitude is kept constant at the maximum input voltage range of the ADC over the measured frequency range. The losses of the hybrid, the RF cables and the RF substrate are compensated. The digital output data is either stored in the FPGA or captured by a four channel 40 GS/s real-time scope. The measurement data is transferred to a PC where a discrete Fourier transform (DFT) is computed by means of Matlab to calculate the SNDR.

The real-time scope provides only four measurement channels. Thus only one of the four ADC channels can be measured simultaneously. The advantage of the FPGA measurement is that the twelve output channels of the four sub-ADCs can be measured simultaneously. The drawback of the current FPGA design is that the data transmission speed is limited to 3.2 Gbit/s per channel. This limits the sampling rate of the ADC to 12.8 GS/s. The FPGA measurement environment is explained in detail in [7].

1) *FPGA Measurement Results:* In Fig. 6 the single channel measurement results for a combined sampling rate of 12.8 GS/s are shown. Due to mismatch the results of the individual channels differ by up to 3 dB. All channels have a SNDR above 15 dB up to 15 GHz.

In Fig. 7 the SNDR of the ADC is shown after multiplexing the digital output data by means of Matlab and subsequent calculation of the SNDR. The resulting SNDR is about 15 dB up to 15 GHz input signal frequency. This corresponds to a resolution of at least 2 bit.

2) *Real-Time Scope Measurement Results:* Figure 8 shows the single channel measurement result for a single channel sampling rate of 6.4 GS/s. The increased sampling frequency doesn't decrease the SNDR compared to the FPGA measurement results at 3.2 GS/s. A further increasing of the sampling rate to 9 GS/s per channel (cf. Fig. 9) reduces the SNDR by about 2 dB between 12 GHz and 16 GHz input signal frequency in comparison to the measurement at the lower sampling rate.

Due to a limited number of scope channels only one ADC channel and the most significant bit (MSB) of the adjacent

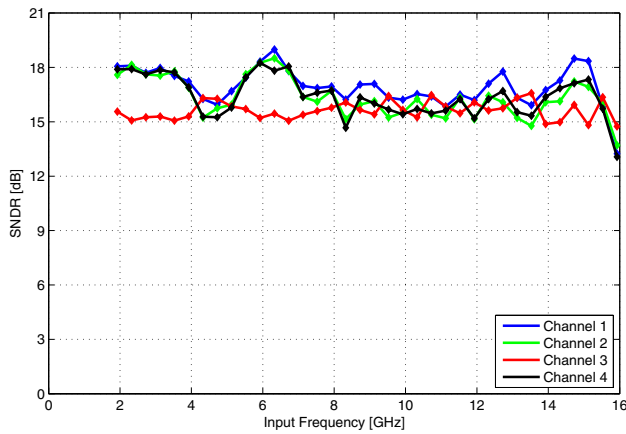


Fig. 6. Signal to noise ratio versus the input signal frequency of the four ADC channels, each channel is operating at 3.2 GS/s.

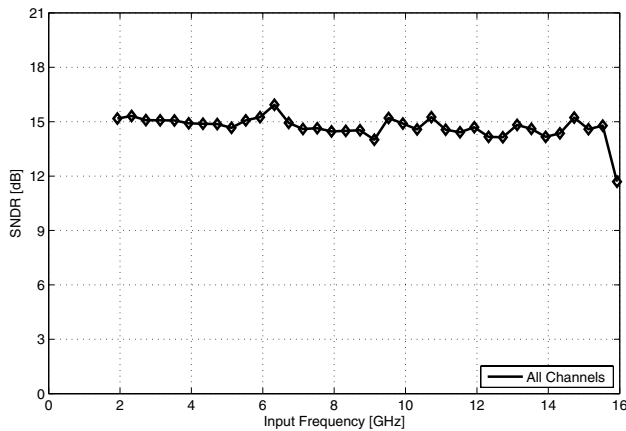


Fig. 7. SNDR of the multiplexed output at 12.8 GS/s versus the input signal frequency of the ADC.

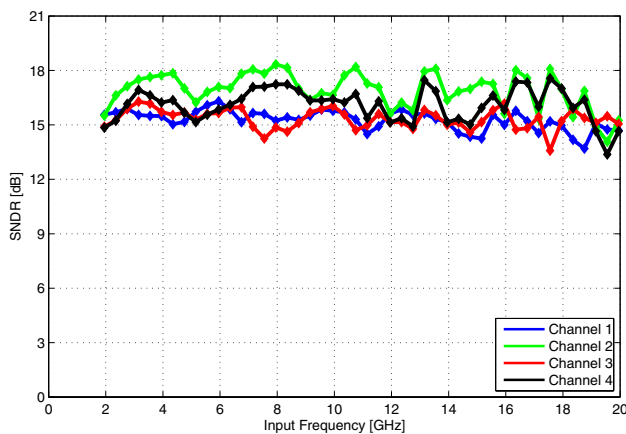


Fig. 8. Signal to noise ratio versus the input signal frequency of the four ADC channels, each channel is operating at 6.4 GS/s.

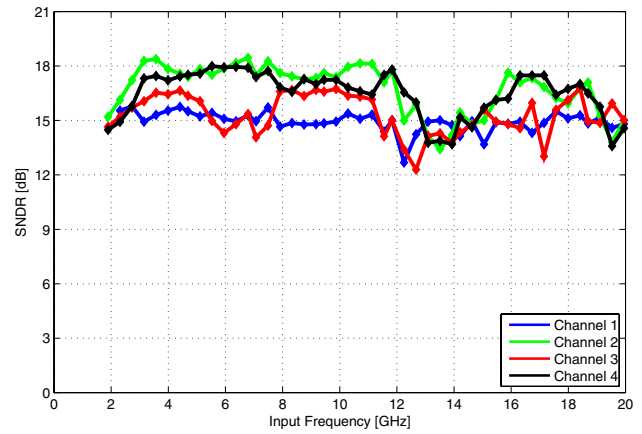


Fig. 9. Signal to noise ratio versus the input signal frequency of the four ADC channels, each channel is operating at 9 GS/s.

ADC channel can be measured simultaneously. To calculate the SNDR of the time interleaved ADC (TIADC) from the single channel measurements, it is necessary to take offset errors, gain errors and timing mismatches into account [8].

The calculation of the offset error is performed in the frequency domain by calculating the DFT for each ADC channel. The resulting frequency spectrum gives also the DC signal power for each channel. The offset mismatch is now calculated as the deviation of each channels DC component from the cumulative mean of all DC components. This way, the offset error can be incorporated into the SNDR calculation in the frequency domain.

According to the offset calculation, the gain error is calculated as the deviation of each channels signal amplitude from the mean value of all channels amplitudes. This way, the power at the frequency bin of the DFT associated with the input signal can be separated into signal power and power resulting from the gain error.

The timing mismatch calculation is more sophisticated, as it has to be estimated from the measurement of the MSB of the adjacent ADC channel. The calculation of the timing mismatch is performed in the time domain. The input phase of a sinusoid input signal is varied until the mean error between the input signal and the measured signal is minimized. This is done with the measured channel and with the MSB of the adjacent channel. The limited resolution of the ADC leads to a limited accuracy of this estimation.

The difference in the estimated input signal phases allows to calculate the timing deviation between the channels, which in turn leads to a worst-case estimation of the associated error power that adds to the whole error power when calculating the SNDR of the TIADC. In Fig. 10 the mean timing mismatch at different ADC input signal frequencies is shown for all channels. The worst case timing deviation is 0.1 T_s , the mean timing deviation is about 0.05 T_s , where T_s is the sampling period of the TIADC.

By taking the estimated offset errors, gain errors and timing mismatch into account, the SNDR of the TIADC in Fig. 11

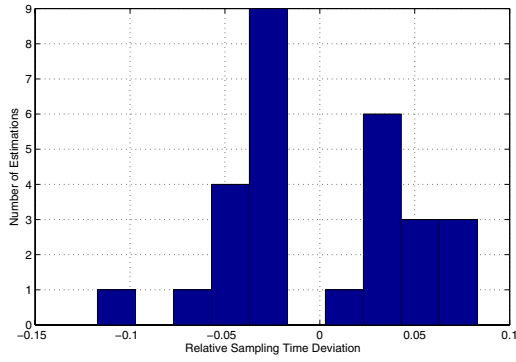


Fig. 10. Distribution of the estimated sampling time deviation, normalized with the sampling time T_s of the TIADC.

is calculated. The topmost curve is the SNDR without timing mismatch. With a timing deviation of $0.05 T_s$ the resolution is reduced marginal compared to an ADC without timing deviation. The lowermost curve is a worst case estimation with a timing deviation of $0.1 T_s$, this reduces the SNDR by up to 5 dB for very high input signal frequencies.

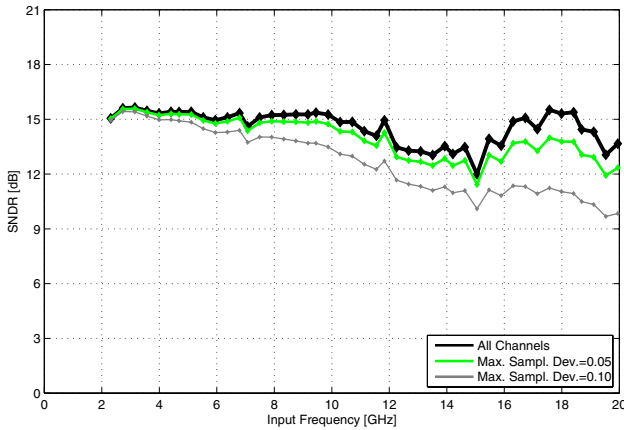


Fig. 11. The SNDR versus the input signal frequency of the TIADC at 36 GS/s, without timing deviation and with a timing deviation of $0.05 T_s$ and $0.1 T_s$. Offset and gain errors are included in all curves.

The algorithms and the complete procedure is verified in Matlab with a modeled TIADC which exhibits offset errors, gain errors and timing mismatch.

IV. COMPARISON BETWEEN STATE-OF-THE-ART ADCs

Table I shows some representative state-of-the-art ADCs. The converters [9] and [10] are intended for the same application as the ADC in this work. Converter [3] is the ADC with the highest resolution and sampling rate available.

When comparing this work with the SiGe converters [9], [10] it is obvious that CMOS has the same performance with a lower power consumption. When the power saving principles that we used in [11] are applied to this design, the power consumption of this design can be reduced by a factor of 10 to about 250 mW. Another advantage of CMOS is the ability

TABLE I
HIGH SPEED STATE-OF-THE-ART ADCs

Vendor	[9]	[10]	[3]	This work
Technology	SiGe	SiGe BiCMOS	CMOS	CMOS
Publication	2004	2010	2010	
Sampling rate	40 GS/s	40 GS/s	56 GS/s	36 GS/s
Nom. Resolution	3	4	8	3
ENOB@15 GHz	2 bit	3 bit	6 bit	2 bit
Core Power	3.8 W	5.5 W	2 W	2.6 W
Core Size	1.8 mm ²	1.4 mm ²	16 mm ²	0.16 mm ²

to integrate the ADC and a digital signal processor on the same chip. This is a large cost benefit compared to the multi chip solution. Also the core area is about 10 times smaller than in SiGe Technology [9], [10].

Comparing this work with the converter in [3] is difficult, because the applications are different. The advantage of the three bit ADC for the application in a digital equalizer is the small core size, the lower price of the IP due to a lower complexity and a potentially lower power consumption after rescaling of the ADC components.

For the application together with a digital equalizer the presented 3 bit ADC is the best choice concerning the chip area, the power consumption, the performance and the overall system costs.

V. CONCLUSION

The presented 3 bit 36 GS/s CMOS ADC achieves an effective resolution of 2 bit, while occupying a very small core area of 0.16 mm². This is the smallest chip area ever reported for an ADC with a sampling rate of 36 GS/s, and a hundred times smaller than the currently fastest CMOS ADC [3]. This allows for the cost-effective integration with a digital equalizer for high speed fiber-optic systems on a single CMOS chip [6].

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